Numerical Analysis Assignment 3

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Assessed exercise 3.1 (Stability analysis of the scalar linear IVP) Let Y be the solution of the scalar first order linear IVP:

$$\begin{aligned} \dot{y} &= f(t,y) := a(t)\, y, \\ y(0) &= y_0, \end{aligned} \tag{1}$$

where $a \in C^0(\mathbb{R})$ is a continuous function satisfying

$$\lambda := \sup_{t\in\mathbb{R}} |a(t)| < \infty.$$

Let $\epsilon > 0$ and suppose that $b \in C^0(\mathbb{R})$ and $\delta \in \mathbb{R}$ are such that

$$\|b\|_{L^\infty} \leq \epsilon, \qquad |\delta| \leq \epsilon.$$

Consider the solution $\tilde{Y}$ of the perturbed IVP

$$\begin{aligned} \dot{y} &= \tilde{f}(t,y) := a(t)\, y + b(t), \\ y(0) &= y_0 + \delta. \end{aligned} \tag{2}$$

1. By showing that both f and $\tilde{f}$ are globally Lipschitz in the appropriate sense, use the results from lectures to deduce that both (1) and (2) have unique solutions. Furthermore, what is the Lipschitz constant?

SOLUTION:

To show that f and $\tilde{f}$ are globally Lipschitz in the second variable we have:

$$|f(t,y_1) - f(t,y_2)| = |a(t)y_1 - a(t)y_2| = |a(t)|\,|y_1 - y_2| \leq \lambda\,|y_1 - y_2|$$

Therefore f is globablly Lipschitz in the second variable on its entire domain (we could denote this as D) with Lipschitz constant $L = \lambda$.

Similarly for $\tilde{f}$ we have:

$$|\tilde{f}(t,y_1) - \tilde{f}(t,y_2)| = |a(t)y_1 + b(t) - (a(t)y_2 + b(t))| = |a(t)|\,|y_1 - y_2| \leq \lambda\,|y_1 - y_2|$$

Therefore $\tilde{f}$ is also globally Lipschitz in the second variable on D with the same Lipschitz constant $L = \lambda$.

Now in order to show that both IVPs have unique solutions we notice that since a and b are continuous, and f and $\tilde{f}$ are linear/affine in y, we have that both f and $\tilde{f}$ are continuous on D.

Now since our Lipschitz inequalities hold for all $(t,y_1),(t,y_2) \in D$, we have that, independently of $(t_0,y_0) \in D$, it holds that $\forall\, \rho > 0$ and all $(t,y_1),(t,y_2) \in \overline{B_\rho((t_0,y_0))} \cap D$:

$$|f(t,y_1) - f(t,y_2)| \leq \lambda |y_1 - y_2|$$

I.e. regardless of initial conditions and radius ρ, then for any points in the ball centered at the initial conditions, (t_0,y_0) with radius ρ the local Lipschitz condition holds with $K_0 = \lambda$. (The same applies for $\tilde{f}$).

Therefore, by Theorem 5.8 both IVPs admit a unique local solution: Y (and $\tilde{Y}$), on an open interval I (and $\tilde{I}$) of the form $(t_0-\delta, t_0+\delta)$ such that $Y : I \to \mathbb{R}$ (and $\tilde{Y} : \tilde{I} \to \mathbb{R}$).

Now, by Proposition 5.13, we have that by taking larger neighbourhoods about the respective initial conditions (i.e. increasing $\rho \implies$ increasing $\delta \implies$ larger time

intervals), the solutions on the overlapping intervals must still coincide on their intersections. Therefore we can obtain maximal solutions:

$$Y_{\max} : I_{\max} \to \mathbb{R} \quad \text{and} \quad \tilde{Y}_{\max} : \tilde{I}_{\max} \to \mathbb{R}$$

Then since we have shown f and $\tilde{f}$ are globally Lipschitz, it follows by theorem 5.16 that:

$$I_{\max} = \mathbb{R}, \qquad \tilde{I}_{\max} = \mathbb{R}$$

Hence both of the IVPs have a unique solution defined on the entirety of $\mathbb{R}$.

2. Show that the perturbation error $e(t) := \tilde{Y}(t) - Y(t)$ satisfies

$$\begin{aligned} e'(t) &= a(t)\, e(t) + b(t), \\ e(0) &= \delta. \end{aligned}$$

SOLUTION:

We have:

$$\begin{aligned} e'(t) &= \tilde{Y}'(t) - Y'(t) \\ &= \Big(a(t)\, \tilde{Y}(t) + b(t)\Big) - a(t)Y(t) \\ &= a(t)\left(\tilde{Y}(t) - Y(t)\right) + b(t) \\ &= a(t)e(t) + b(t) \end{aligned}$$

Then:

$$e(0) = \tilde{Y}(0) - Y(0) = (y_0 + \delta) - y_0 = \delta$$

3. Using an integrating factor or otherwise, show that

$$e(t) = \delta\, \exp(A(t)) + \int_0^t b(s)\, \exp\left(A(t) - A(s)\right)\, ds, \tag{3}$$

where

$$A(t) := \int_0^t a(s)\, ds. \tag{4}$$

SOLUTION:

We are given:

$$A(t) := \int_0^t a(s)\, ds \implies A'(t) = a(t)$$

As we have shown:

$$e'(t) = a(t)e(t) + b(t)$$

We can multiply through by a to be determined integrating factor, μ:

$$\implies e'(t)\mu(t) = a(t)\mu(t)e(t) + \mu(t)b(t)$$

This suggests that we want our $\mu(t)$ to be such that $a(t)\mu(t) = -\mu'(t)$ to ensure we can then apply the product rule.

$$a(t)\mu(t) = -\mu'(t) \implies \mu(t) = \exp\left(-\int_0^t a(s)\, ds\right) = \exp(-A(t))$$

Hence we have:

$$\begin{aligned} & e'(t)\mu(t) = a(t)\mu(t)e(t) + \mu(t)b(t) \\ \implies\ & \mu(t)b(t) = e'(t)\mu(t) - a(t)\mu(t)e(t) \\ & \qquad = e'(t)\mu(t) - a(t)\exp(-A(t))e(t) \\ & \qquad = e'(t)\mu(t) + \left(\exp(-A(t))\right)' e(t) \\ & \qquad = e'(t)\mu(t) + \left(\mu(t)\right)' e(t) \end{aligned}$$

$$= (e(t)\mu(t))'$$

So we have:

$$(e(t)\exp(-A(t)))' = b(t)\exp(-A(t))$$

Integrating both sides from 0 to t yields:

$$\begin{aligned} e(t)\exp(-A(t)) - e(0)\exp(-A(0)) &= \int_0^t b(s)\exp(-A(s))\,ds \\ \implies e(t)\exp(-A(t)) - \delta &= \int_0^t b(s)\exp(-A(s))\,ds \\ \implies e(t) &= \delta\exp(A(t)) + \int_0^t b(s)\exp(A(t)-A(s))\,ds \end{aligned}$$

4. Suppose that there is $\mu \leq \lambda$ such that $a(t) \in [-\lambda, \mu]$ for all $t \in \mathbb{R}$.

   (a) Show that if $\mu < 0$, then

$$|e(t)| \;\leq\; \frac{1 + (|\mu| - 1)\,\exp(\mu t)}{|\mu|}\,\epsilon \;\leq\; \begin{cases} \epsilon/|\mu|, & \mu \in (-1, 0), \\ \epsilon, & \mu \leq -1, \end{cases} \tag{5}$$

   and hence, as $\epsilon \to 0$, the perturbation error converges to zero uniformly on $[0, \infty)$.

SOLUTION:

As we saw above:

$$e(t) = \delta\exp(A(t)) + \int_0^t b(s)\exp(A(t)-A(s))\,ds$$

We therefore have:

$$\begin{aligned} |e(t)| &= \left|\delta\exp(A(t)) + \int_0^t b(s)\exp(A(t)-A(s))\,ds\right| \\ &\leq |\delta\exp(A(t))| + \left|\int_0^t b(s)\exp(A(t)-A(s))\,ds\right| \end{aligned}$$

Then since we are given that $|\delta| \leq \epsilon$ and $\|b\|_{L^\infty} \leq \epsilon$, we have:

$$\begin{aligned} |e(t)| &\leq \epsilon\exp(A(t)) + \left|\int_0^t b(s)\exp(A(t)-A(s))\,ds\right| \\ &\leq \epsilon\exp(A(t)) + \epsilon\left|\int_0^t \exp(A(t)-A(s))\,ds\right| \end{aligned}$$

Since $a(t) \in [-\lambda, \mu]$ we have:

$$A(t) = \int_0^t a(s)\,ds \leq \int_0^t \mu\,ds = \mu t \implies \exp(A(t)) \leq \exp(\mu t)$$

Therefore the previous inequality becomes:

$$|e(t)| \leq \epsilon\exp(\mu t) + \epsilon\left|\int_0^t \exp(A(t)-A(s))\,ds\right|$$

Considering $\left|\int_0^t \exp(A(t)-A(s))\,ds\right|$ we have:

$$\begin{aligned} \left|\int_0^t \exp(A(t)-A(s))\,ds\right| &= \left|\int_0^t \exp\left(\int_s^t a(u)\,du\right)\,ds\right| \\ &\leq \left|\int_0^t \exp\left(\int_s^t \mu\,du\right)\,ds\right| \end{aligned}$$

$$
\begin{aligned}
&= \left| \int_0^t \exp\left(\mu(t-s)\right)\, ds \right| \\
&= \left| \int_0^t \exp(\mu v)\, dv \right| \\
&= \left| \frac{1}{\mu} \left(\exp(\mu t) - 1\right) \right|
\end{aligned}
$$

Given we have that $\mu < 0 \implies \exp(\mu t) \leq 1$ (for $t \geq 0$) $\implies \exp(\mu t) - 1 \leq 0$ therefore:

$$
\left| \int_0^t \exp(A(t) - A(s))\, ds \right| \leq \left| \frac{\exp(\mu t) - 1}{\mu} \right| = \frac{1 - \exp(\mu t)}{|\mu|}
$$

Therefore we have:

$$
\begin{aligned}
|e(t)| &\leq \epsilon \exp(\mu t) + \epsilon \left| \int_0^t \exp(A(t) - A(s))\, ds \right| \\
&\leq \epsilon \exp(\mu t) + \epsilon \frac{1 - \exp(\mu t)}{|\mu|} \\
&= \epsilon \left(\exp(\mu t) + \frac{1 - \exp(\mu t)}{|\mu|} \right) \\
&= \epsilon \left(\frac{|\mu| \exp(\mu t) + 1 - \exp(\mu t)}{|\mu|} \right) \\
&= \epsilon \left(\frac{\exp(\mu t)\left(|\mu| - 1\right) + 1}{|\mu|} \right)
\end{aligned}
$$

Then if $\mu \in (-1, 0)$ we have:

$$
\begin{aligned}
\mu \in (-1,0) &\implies |\mu| \in (0,1) \\
&\implies |\mu| - 1 \in (-1,0) \\
\text{Also } \mu \in (-1,0) &\implies \exp(\mu t) \in (0,1] \\
&\implies \left(|\mu| - 1\right) \exp(\mu t) \in (-1,0) \\
&\implies \left(|\mu| - 1\right) \exp(\mu t) + 1 \in (0,1) \\
&\implies \left[\left(|\mu| - 1\right) \exp(\mu t) + 1\right] \frac{\epsilon}{|\mu|} \in (0, \frac{\epsilon}{|\mu|})
\end{aligned}
$$

Or, if $\mu \leq -1$ we have:

$$
\mu \leq -1 \implies |\mu| \geq 1 \implies |\mu| - 1 \geq 0
$$

$$
\begin{aligned}
\text{Also } \mu \leq -1 &\implies \exp(\mu t) \in (0,1] \\
&\implies \exp(\mu t)\left(|\mu| - 1\right) \leq |\mu| - 1 \\
&\implies 1 + \exp(\mu t)\left(|\mu| - 1\right) \leq |\mu| \\
&\implies \frac{1 + \exp(\mu t)\left(|\mu| - 1\right)}{|\mu|} \leq 1 \\
&\implies \frac{1 + \exp(\mu t)\left(|\mu| - 1\right)}{|\mu|} \epsilon \leq \epsilon
\end{aligned}
$$

Therefore:

$$
|e(t)| \leq \frac{1 + \left(|\mu| - 1\right) \exp(\mu t)}{|\mu|} \epsilon \leq \begin{cases} \epsilon/|\mu|, & \mu \in (-1, 0), \\ \epsilon, & \mu \leq -1 \end{cases}
$$

Now to show that $e(t)$ converges to zero uniformly on $[0, \infty)$ we can define:

$$
C_\mu := \begin{cases} \frac{1}{|\mu|}, & \mu \in (-1, 0), \\ 1, & \mu \leq -1. \end{cases}
$$

Where we notice that $C_\mu > 0$ is independent of t and ϵ. Then it holds from the previous result that $\forall t \in [0, \infty)$:

$$|e(t)| \leq \frac{1 + (|\mu| - 1)\, \exp(\mu t)}{|\mu|}\, \epsilon \leq C_\mu \epsilon$$

Then because C_μ is independent of t, it holds that:

$$\sup_{t \in [0,\infty)} |e(t)| \leq C_\mu\, \epsilon,$$

Therefore we have:

$$\lim_{\epsilon \to 0} \sup_{t \in [0,\infty)} |e(t)| \;\leq\; \lim_{\epsilon \to 0} C_\mu\, \epsilon = C_\mu \lim_{\epsilon \to 0} \epsilon = 0.$$

Therefore it holds that as $\epsilon \to 0$, the perturbation error converges to zero uniformly on $[0, \infty)$.

(b) Show that if $\mu > 0$, then

$$|e(t)| \;\leq\; \frac{(\mu + 1)\, \exp(\mu t) - 1}{\mu}\, \epsilon. \tag{6}$$

SOLUTION:
As we have seen previously:

$$\begin{aligned}
e(t) &= \delta \exp(A(t)) + \int_0^t b(s) \exp(A(t) - A(s))\, ds \\
\implies |e(t)| &\leq \epsilon \exp(A(t)) + \epsilon \left| \int_0^t \exp(A(t) - A(s))\, ds \right| \\
&\leq \epsilon \exp(\mu t) + \epsilon \left| \frac{1}{\mu} \left(\exp(\mu t) - 1 \right) \right| \\
&= \epsilon \left[\frac{\exp(\mu t)\mu}{\mu} + \frac{\exp(\mu t) - 1}{\mu} \right] \\
&= \epsilon \left[\frac{\exp(\mu t)\, (\mu + 1) - 1}{\mu} \right]
\end{aligned}$$

As $\epsilon \to 0$ can it be concluded that

- $e(t)$ converges to zero for each fixed t?
- e converges to zero uniformly on any bounded time interval $[0, T]$ ($T < \infty$)?
- e converges to zero uniformly on $[0, \infty)$?

SOLUTION:

We have shown that for $\mu > 0$ it holds that:

$$|e(t)| \leq \frac{(\mu + 1)\, \exp(\mu t) - 1}{\mu}\, \epsilon$$

Then for each fixed t, we can define C_t:

$$C_t := \frac{(\mu + 1)\, \exp(\mu t) - 1}{\mu}$$

Where for each fixed t, C_t is finite. Hence we have that that for a fixed t:

$$|e(t)| \leq C_t \epsilon$$

$$\implies \lim_{\epsilon \to 0} |e(t)| \leq \lim_{\epsilon \to 0} C_t \epsilon = C_t \lim_{\epsilon \to 0} \epsilon = 0$$

Hence, $e(t)$ converges to 0 for each fixed t.

On a bounded time interval $[0, T]$ with $T < \infty$ we have that since C_t is strictly increasing in t:

$$\sup_{t\in[0,T]} C_t = C(T) = \frac{(\mu+1)\,\exp(\mu T) - 1}{\mu}$$

Therefore we have that for any bounded time interval, $[0, T]$:

$$\forall t \in [0, T] \,:\, |e(t)| \leq C_t\,\epsilon \leq C(T)\,\epsilon$$

$$\implies \lim_{\epsilon\to 0} \sup_{t\in[0,T]} |e(t)| \leq \lim_{\epsilon\to 0} \sup_{t\in[0,T]} C_t\epsilon = \lim_{\epsilon\to 0} C(T)\,\epsilon = C(T) \lim_{\epsilon\to 0} \epsilon = 0$$

Therefore $e(t)$ converges to zero uniformly on any bounded interval $[0, T]$.

If we have an unbounded interval (i.e. $T = \infty$), then:

$$(C(\infty) =) \lim_{t\to\infty} C_t = \lim_{t\to\infty} \frac{(\mu+1)\,\exp(\mu t) - 1}{\mu} = \infty$$

$$\implies \sup_{t\in[0,\infty)} C_t = \infty$$

Therefore we cannot conclude that:

$$\lim_{\epsilon\to 0} \sup_{t\in[0,\infty)} C_t\epsilon = 0$$

I.e. we cannot conclude that e converges to 0 uniformly on $[0, \infty)$.

Assessed exercise 3.2 (Second order ODEs) Consider the ODE

$$\begin{aligned} \ddot{y} + \sin(\dot{y}) &= \cos(\omega t), \\ \dot{y}(0) &= a, \\ y(0) &= b \end{aligned}$$

for $a, b, \omega \in \mathbb{R}$.

Supposing sufficient regularity holds, we will show that there exists a solution to the above second order ODE.

1. Construct a vector–valued first order system of the form

$$(\dot{y}, \dot{z}) = f\left(t, (y, z)\right)$$

such that Y which solves the second order system should be (the first component of) the solution of this system, where the function f should be stated along with appropriate initial conditions.

SOLUTION:

Defining:

$$z(t) := \dot{y}(t)$$

We then have:

$$\dot{z}(t) = \ddot{y}(t)$$

So our existing ODE:

$$\ddot{y} + \sin(\dot{y}) = \cos(\omega t)$$

Becomes:

$$\dot{z}(t) + \sin(z(t)) = \cos(\omega t) \implies \dot{z}(t) = \cos(\omega t) - \sin(z(t))$$

Therefore, we now have the vector valued first order system of the form:

$$\begin{aligned} \dot{y}(t) &= z(t) \\ \dot{z}(t) &= \cos(\omega t) - \sin\left(z(t)\right) \end{aligned}$$

This can be expressed in the required form:

$$(\dot{y}(t), \dot{z}(t)) = f\left(t, (y(t), z(t))\right)$$

Where we have that:

$$f\left(t, (y, z)\right) := (z(t), \cos(\omega t) - \sin(z))$$

Our initial conditions then become:

$$y(0) = b \text{ as before, and } z(0) = \dot{y}(0) = a$$

So altogether we have:

$$\begin{aligned} \dot{y}(t) &= z(t) \\ \dot{z}(t) &= \cos(\omega t) - \sin\left(z(t)\right) \\ y(0) &= b \\ z(0) &= a \end{aligned}$$

2. Show that the f constructed above is appropriately smooth, hence apply a result from lecture notes in order to conclude the unique existence (at least for small time) of a pair (Y, Z) which solves the first order system.

SOLUTION:

Here we want to apply Theorem 5.8 which requires that f is continiuous, and is Lipschitz in the second variable. We have that:

$$f\left(t, (y, z)\right) := (z, \cos(\omega t) - \sin(z))$$

We can take that since both components of f are made from linear combinations of smooth functions: z, $\cos(\omega t)$ and $\sin(z)$, it holds that f is C^∞. To show then that we have f being Lipschitz in the second variable, we can define:

$$u_1 = (y_1, z_1), \quad u_2 = (y_2, z_2)$$

Then we have:

$$\begin{aligned}
\|f(t,(y_1,z_1)) - f(t,(y_2,z_2))\| &= \|f(t,u_1) - f(t,u_2)\| \\
&= \|(z_1, \cos(\omega t) - \sin(z_1)) - (z_2, \cos(\omega t) - \sin(z_2))\| \\
&= \|(z_1 - z_2, \sin(z_2) - \sin(z_1))\| \\
&= \sqrt{(z_1 - z_2)^2 + (\sin(z_2) - \sin(z_1))^2}
\end{aligned}$$

At this point we can notice that by the Mean Value Theorem that there must exist some $\tilde{z}$ between z_1 and z_2 such that:

$$\sin(z_1) - \sin(z_2) = \sin'(\tilde{z})\,(z_1 - z_2) = \cos(\tilde{z})\,(z_1 - z_2).$$

$$\implies |\sin(z_1) - \sin(z_2)| = |\cos(\tilde{z})||z_1 - z_2| \leq |z_1 - z_2|$$

$$\implies (\sin(z_2) - \sin(z_1))^2 \leq (z_1 - z_2)^2$$

Hence sin is Lipschitz with constant 1. So our expression becomes:

$$\begin{aligned}
\|f(t,u_1) - f(t,u_2)\| &= \sqrt{(z_1 - z_2)^2 + (\sin(z_2) - \sin(z_1))^2} \\
&\leq \sqrt{(z_1 - z_2)^2 + (z_1 - z_2)^2} \\
&= \sqrt{2}\sqrt{(z_1 - z_2)^2} \\
&\leq \sqrt{2}\sqrt{(z_1 - z_2)^2 + (y_1 - y_2)^2} \\
&= \sqrt{2}\|u_1 - u_2\|
\end{aligned}$$

Hence f is indeed Lipschitz in the second variable, $(y, z) = u$ (with Lipschitz constant $\sqrt{2}$).

Then applying Theorem 5.8 immediatetly then gives us that around the initial conditions given:

$$(t_0, (y(0), z(0))) = (0, u(0)) = (0, (b, a))$$

There exists some $\delta > 0$ whereby there is a unique local solution $u(t) = (Y(t), Z(t))$ defined locally on $(-\delta, \delta)$ such that:

$$\dot{u}(t) = f\left(t, u(t)\right), \qquad u(0) = (b, a).$$

I.e. there exists a unique pair (Y, Z), at least for small time, which solves the first order system. Note that since f is globally Lipschitz in its second variable, we could actually apply the same kind of result obtained in 3.1.1 to state that rather than just for a small time, we actually have the same unique solution for all time.

Assessed exercise 3.3 Consider the simple ODE

$$\dot{y} = \lambda y, \qquad y(0) = 1,$$

where $\lambda > 0$ is some given number. This has solution $Y(x) = \exp(\lambda x)$.

1. State the forward Euler discretisation of the above ODE with N points and a uniform discretisation on $[0, 1]$.

SOLUTION:

Discretising the interval $[0, 1]$ uniformly into N points gives us that:

$$h = \frac{1}{N}, \qquad x_n := nh, \quad n = 0, 1, \ldots, N$$

We can then give the FE discretisation of the ODE as:

$$\begin{cases} y^{n+1} = y^n + \frac{1}{N}\lambda y^n & = y^n\left(1 + \frac{\lambda}{N}\right) \\ y^0 & = 1 \end{cases} ; \, n = 0, 1, \ldots, N-1$$

2. Solve the forward Euler discretisation (this is possible by hand) to obtain an explicit form of y^n which should approximate $Y(x_n)$.

SOLUTION:

Evaluating the first few iterations yields:

$$\begin{aligned} y^1 &= y^0(1 + \frac{\lambda}{N}) = (1 + \frac{\lambda}{N}) \\ y^2 &= (1 + \frac{\lambda}{N})\, y^1 = (1 + \frac{\lambda}{N})^2 \\ &\vdots \\ y^n &= (1 + \frac{\lambda}{N})^n \; : \qquad n = 0, 1, \ldots, N \end{aligned}$$

We can reexpress this as:

$$y^n = (1 + \frac{\lambda}{N})^{x_n N} = \left[\left(1 + \frac{\lambda}{N}\right)^N\right]^{x_n}$$

For large N, we therefore have:

$$y^n = \left[\left(1 + \frac{\lambda}{N}\right)^N\right]^{x_n} \approx \exp(\lambda x_n)$$

Hence we have that y^n approximates the exact solution $Y(x) = \exp(\lambda x)$ on the discretised set of points along the interval $[0, 1]$.

3. Suppose that $(1 + h\lambda)^n = \exp(\lambda nh)\left(1 - \frac{\lambda^2 nh^2}{2}\right) + \mathcal{O}(h^3)$. Show that there exists $c > 0$ such that for all h sufficiently small, $Y(1) - y^N \geq ch$.

SOLUTION:

Since we know that the solution $Y(x) = \exp(\lambda x)$, it holds that $Y(1) = \exp(\lambda)$.

Therefore:

$$Y(1) - y^N = \exp(\lambda) - y^N$$

Then since we have from the FE discretisation:

$$y^n = \left(1 + \frac{\lambda}{N}\right)^n = (1 + h\lambda)^n \implies y^N = (1 + h\lambda)^N$$

Applying the equality provided:

$$(1+h\lambda)^n = \exp(\lambda nh)\left(1-\frac{\lambda^2 nh^2}{2}\right)+\mathcal{O}(h^3)$$

We have for $n=N$:

$$y^N = (1+h\lambda)^N = \exp(\lambda Nh)\left(1-\frac{\lambda^2 Nh^2}{2}\right)+\mathcal{O}(h^3)$$

Then since it holds that $Nh=1$:

$$y^N = (1+h\lambda)^N = \exp(\lambda)\left(1-\frac{\lambda^2 h}{2}\right)+\mathcal{O}(h^3)$$

$$\begin{aligned}\implies Y(1)-y^N &= \exp(\lambda)-\exp(\lambda)\left(1-\frac{\lambda^2 h}{2}\right)-\mathcal{O}(h^3)\\ &= \exp(\lambda)\left(\frac{\lambda^2 h}{2}\right)-\mathcal{O}(h^3)\\ &= \exp(\lambda)\left(\frac{\lambda^2 h}{2}\right)+\mathcal{O}(h^3) \text{ by convention of big-O notation}\end{aligned}$$

At this stage we can define the constant term:

$$C := \frac{\lambda^2}{2}\exp(\lambda) > 0$$

So we can then say:

$$Y(1)-y^N = Ch+\mathcal{O}(h^3)$$

Since we know that:

$$\lim_{h\to 0}\frac{\mathcal{O}(h^3)}{h}=0 \implies \exists h_C^* \text{ such that } \forall h\in(0,h_C^*):$$

$$\left|\frac{\mathcal{O}(h^3)}{h}\right| \le \frac{C}{2} \implies \left|\mathcal{O}(h^3)\right| \le \frac{C}{2}h$$

Therefore:

$$\forall h\in(0,h_C^*):\ Y(1)-y^N = Ch+\mathcal{O}(h^3) \ \ge\ Ch-\left|\mathcal{O}(h^3)\right| \ \ge\ Ch-\frac{C}{2}h=\frac{C}{2}h$$

Hence $\forall\, h\in(0,h_C^*)\ :\ Y(1)-y^N \ge \frac{C}{2}h$ and so taking $c:=\frac{C}{2}$ it holds that for all h sufficiently small we have that:

$$Y(1)-y^N \ge ch$$

For **fun, but no marks**, show that $(1+h\lambda)^n = \exp(\lambda nh)\left(1-\frac{\lambda^2 nh^2}{2}\right)+\mathcal{O}(h^3)$.

SOLUTION:

This is a matter of applying a Binomial and Taylor expansion respectively to LHS and RHS.

- We apply the Binomial Theorem to LHS: $(1+h\lambda)^n$

$$\begin{aligned}(1+h\lambda)^n &= \sum_{k=0}^{n}\binom{n}{k}(\lambda h)^k = 1+n\cdot\lambda h+\frac{n(n-1)}{2}\lambda^2h^2+\sum_{k=3}^{n}\binom{n}{k}(\lambda h)^k\\ &= 1+n\cdot\lambda h+\frac{n(n-1)}{2}\lambda^2h^2+\mathcal{O}(h^3)\\ &= 1+\lambda nh+\frac{\lambda^2 n(n-1)}{2}h^2+\mathcal{O}(h^3)\end{aligned}$$

- We then apply the Taylor expansion of e^x to RHS : $\exp(\lambda nh)$

$$\begin{aligned}\exp(\lambda nh) &= 1 + \lambda nh + \frac{\lambda^2 n^2 h^2}{2} + \frac{(\lambda nh)^k}{k!} \\ &= 1 + \lambda nh + \frac{\lambda^2 n^2 h^2}{2} + \mathcal{O}(h^3)\end{aligned}$$

Multiplying through by $\left(1 - \frac{\lambda^2 nh^2}{2}\right)$ yields:

$$\begin{aligned}\exp(\lambda nh)\left(1 - \frac{\lambda^2 nh^2}{2}\right) &= \left(1 + \lambda nh + \frac{\lambda^2 n^2 h^2}{2} + \mathcal{O}(h^3)\right)\left(1 - \frac{\lambda^2 nh^2}{2}\right) \\ &= 1 + \lambda nh + \frac{\lambda^2 n^2 h^2}{2} - \frac{\lambda^2 nh^2}{2} - \left(\frac{\lambda^2 nh^2}{2}\right)\left[\lambda nh + \frac{\lambda^2 n^2 h^2}{2} + \mathcal{O}(h^3)\right] \\ &= 1 + \lambda nh + \frac{\lambda^2 n^2 h^2}{2} - \frac{\lambda^2 nh^2}{2} + \mathcal{O}(h^3) \\ &= 1 + \lambda nh + \frac{\lambda^2}{2}\left(n^2 - n\right)h^2 + \mathcal{O}(h^3) \\ &= 1 + \lambda nh + \frac{\lambda^2 n(n-1)}{2}h^2 + \mathcal{O}(h^3)\end{aligned}$$

Therefore it holds that

$$(1 + h\lambda)^n = \exp(\lambda nh)\left(1 - \frac{\lambda^2 nh^2}{2}\right) + \mathcal{O}(h^3) \quad \square$$
